SCIENTIFIC COMMITTEE
TWENTY-SECOND REGULAR SESSION

Apia, Samoa
11-19 August 2026

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opal: the open population assessment library

WCPFC-SC22-2026/SA-WP-XX
08 June 2026

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1 Introduction

`[opal](https://n-ducharmebarth-noaa.github.io/opal/)` — the **o**pen **p**opulation **a**ssessment **l**ibrary — is an open-source, modular R package for fisheries stock assessment. The package is built on RTMB (Kristensen, 2023), an R interface to the Template Model Builder framework (Kristensen et al., 2016), which provides automatic differentiation of the model objective function and supports gradient-based optimization and the Laplace approximation for models that include random effects. Writing the model in R rather than C++ keeps the entire model definition within a single language environment, which lowers the barrier to inspecting, modifying, and extending the underlying code.

`opal` builds directly upon the `sbt` R package developed by Quantifish for the assessment of southern bluefin tuna under the Commission for the Conservation of Southern Bluefin Tuna (Webber et al., 2026), and draws on design influences from a number of existing assessment platforms, including the Stochastic Population over Regional Components model (SPoRC; Cheng et al., 2026), the Woods Hole Assessment Model (Miller & Stock, 2020; WHAM; Stock & Miller, 2021), Stock Synthesis (SS3; Methot & Wetzel, 2013), and MULTIFAN-CL (Fournier et al., 1998).

The intent of `opal` is to provide assessment scientists with a flexible modelling framework that supports the estimation of random effects; facilitates enhanced model diagnostics, including one-step-ahead (OSA) residuals, simulation self-testing, and posterior predictive checks; and integrates directly into a Bayesian estimation framework. These capabilities are described in Section 3.

`opal` is being developed under the umbrella of WCPFC Project 123, *Scoping the next generation of tuna stock assessment software* (Magnusson et al., 2024, 2025). Development is conducted as an open collaboration between the Pacific Community (SPC), the Inter-American Tropical Tuna Commission (IATTC), NOAA Fisheries, and other interested parties. The package source code, function documentation, and worked examples are publicly available, and the package is distributed under an open-source license. This paper describes the structure of the model (Section 2), its principal features (Section 3), and two case studies that illustrate its application to a real-data tuna assessment and a simulated single-species assessment (Section 4).

2 Model structure

`opal` estimates age-structured population dynamics and accommodates length-based processes. The population is tracked as numbers-at-age. Biological quantities — weight, maturity, fecundity, natural mortality, and sex ratio — may be supplied on either an age basis or a length basis: a vector defined over age classes is used directly, while a vector defined over length bins is converted to age through a probability-of-length-at-age (PLA) matrix. Selectivity is likewise defined on a length basis and mapped to age through the PLA. This structure allows population demographics to be specified in terms of either age or length, and allows the model to fit multiple data types: relative abundance indices (CPUE), length composition, and weight composition. Where time-varying or hierarchically structured processes are required, parameters may be specified as random effects and integrated out using the Laplace approximation provided by RTMB (Kristensen et al., 2016).

The model is implemented as a single objective function, `opal_model()`, that returns the negative log-likelihood (NLL) for a given set of parameters and data. Rather than encoding the population dynamics and observation model as a single monolithic routine, `opal_model()` orchestrates a sequence of standalone module functions, each responsible for one component of the calculation. The principal modules, in the order they are evaluated, are: mean length-at-age (`get_growth()`); the standard deviation of length-at-age (`get_sd_at_age()`); the PLA matrix (`get_pla()`); weight-, maturity-, and fecundity-at-age (`get_weight_at_length()` , `get_maturity_at_age()` , and a resolver, `resolve_bio_vector()` , that accepts either age- or length-basis inputs); selectivity-at-age (`get_selectivity()` , with the parametric forms `sel_logistic()` and `sel_double_normal()`); initial equilibrium numbers-at-age (`get_initial_numbers()`); the forward population dynamics and harvest (the dynamics routine and `get_harvest_rate()`); recruitment (`get_recruitment()`); the data likelihoods (`get_cpue_like()` , `get_length_like()` , `get_weight_like()`); and the prior and penalty terms (`get_recruitment_prior()` , `evaluate_priors()`). The modular design that this structure reflects is described further in Section 5.1.

2.1 Population dynamics

The population is structured by year, season, and age, and may be configured for any number of fisheries.

2.1.1 Growth and the probability of length-at-age

Mean length-at-age, μ_a, is computed from a Schnute parameterization of the von Bertalanffy growth function, anchored by the lengths L_1 and L_2 at two reference ages A_1 and A_2, which matches the SS3 growth pattern with a linearly interpolated coefficient of variation (Methot & Wetzel, 2013):

$$\mu_a = L_1 + (L_2 - L_1)\,\frac{1 - e^{-k(a-A_1)}}{1 - e^{-k(A_2-A_1)}}.$$

Module: `get_growth()`

The coefficient of variation is interpolated linearly with mean length between the two reference ages, and the standard deviation of length-at-age is its product with the mean:

$$\mathrm{CV}_a = \mathrm{CV}_1 + \frac{\mu_a - L_1}{L_2 - L_1}\,(\mathrm{CV}_2 - \mathrm{CV}_1), \qquad \sigma_a = \mu_a\,\mathrm{CV}_a.$$

Module: `get_sd_at_age()`

From μ_a and σ_a, the probability-of-length-at-age (PLA) matrix gives the probability that an individual of age a falls within length bin ℓ, assuming length-at-age is normally distributed. For contiguous bins with edges $\{e_\ell\}$, this is the difference of the normal cumulative distribution function Φ evaluated at the bin edges:

$$P_{\ell a} = \Phi\left(\frac{e_{\ell+1} - \mu_a}{\sigma_a}\right) - \Phi\left(\frac{e_\ell - \mu_a}{\sigma_a}\right), \qquad \sum_\ell P_{\ell a} = 1.$$

Module: `get_pla()`

The PLA is computed once per function evaluation and reused for the conversion of weight, maturity, fecundity, natural mortality, and selectivity between the length and age bases.

2.1.2 Biology and spawning potential

Quantities defined on a length basis are mapped to age through the PLA. For a length-basis vector x_ℓ, the age-basis vector is $x_a = \sum_\ell P_{\ell a}\, x_\ell$ (equivalently $\mathbf{x}_a = \mathbf{P}^\top \mathbf{x}_\ell$); a vector supplied on an age basis is used directly. Weight-at-length follows an allometric length-weight relationship, $w_\ell = \alpha_{\mathrm{LW}}\, m_\ell^{\beta_{\mathrm{LW}}}$, where m_ℓ is the bin midpoint, and is converted to weight-at-age through the PLA. Spawning potential at age is the product of sex ratio (the fraction mature-female-equivalent at age), maturity, and fecundity:

$$\psi_a = r_a\, m_a\, f_a,$$

Modules: `get_weight_at_length()`, `get_maturity_at_age()`, `resolve_bio_vector()`

where r_a, m_a, and f_a are sex ratio, maturity, and fecundity at age, respectively. Alternatively, a pre-computed spawning-potential vector may be supplied directly. All derived biology arrays are passed explicitly to the dynamics routine rather than read from the data object, so that automatic-differentiation gradients propagate correctly if any of these quantities are estimated (for example, growth parameters estimated through the PLA, or natural mortality specified as a function of length).

2.1.3 Initial conditions

The population is initialized at the unfished equilibrium. Unfished survivorship-per-recruit accumulates natural mortality across ages with a plus-group at the oldest age, $\phi_a^0 = \phi_{a-1}^0\, e^{-M_{a-1}}$ with ϕ_n^0 divided by $1 - e^{-M_n}$, and the unfished spawners-per-recruit is $\Phi_0 = \sum_a \psi_a\, \phi_a^0$. Unfished recruitment and the Beverton-Holt parameters follow from unfished spawning biomass B_0 and steepness h:

$$R_0 = \frac{B_0}{\Phi_0}, \qquad a_{\mathrm{BH}} = \frac{4hR_0}{5h-1}, \qquad b_{\mathrm{BH}} = \frac{B_0(1-h)}{5h-1}.$$

Module: `get_initial_numbers()`

When initial fishing mortality is specified, survivorship is recomputed with total mortality $Z_a = M_a + \sum_f F_f^{\mathrm{init}} s_{f,a}$, giving a fished spawners-per-recruit Φ_{eq} and an equilibrium recruitment $R_{\mathrm{eq}} = a_{\mathrm{BH}} - b_{\mathrm{BH}}/\Phi_{\mathrm{eq}}$. Initial numbers-at-age are $N_a^{\mathrm{init}} = R_{\mathrm{eq}}\, \phi_a$, optionally modified by estimated initial age-structure deviations with the lognormal bias correction described below.

2.1.4 Harvest and the forward projection

The population is projected forward over years and seasons. Within each season, survival from natural mortality is $S_a = e^{-M_a/n_{\mathrm{season}}}$. Removals are taken as a harvest fraction rather than a continuous fishing-mortality rate. For each fishery with observed catch, a fully selected harvest rate is obtained by dividing the observed catch by the available biomass (or available numbers, depending on the catch units of the fishery),

$$F_f = \frac{C_f}{\sum_a N_a\, s_{f,a}\, w_{f,a} + \varepsilon}, \qquad h_{f,a} = F_f\, s_{f,a},$$

Module: `get_harvest_rate()`

where $s_{f,a}$ is selectivity-at-age, $w_{f,a}$ is weight-at-age, and ε is a small constant guarding the denominator. The total harvest rate at age is the sum over fisheries, $h_a = \sum_f h_{f,a}$, and predicted catch-at-age is $\hat{C}_{f,a} = h_{f,a} N_a$. To keep the combined harvest fraction feasible, the quantity $1 - \sum_f F_f$ is passed through a differentiable lower-bound function (`posfun`) that returns the argument where it exceeds a small threshold and otherwise smoothly approaches that threshold, adding the resulting penalty to the objective function.

Surviving numbers are advanced by age and season, with recruitment entering the first age class at the start of each year, and the plus-group accumulating survivors from the oldest two age classes. Spawning biomass is $\mathrm{SB}_y = \sum_a N_{y,a}\, \psi_a$. The routine returns numbers-at-age, predicted catch-at-age by fishery, harvest rates, and the spawning-biomass time series for both the exploited and the dynamic unfished populations; their ratio defines dynamic depletion, $\mathrm{SB}_y / \mathrm{SB}^0_y$.

2.1.5 Recruitment

Recruitment follows a Beverton-Holt stock-recruitment relationship with annual log-normal deviations ε_y:

$$R_y = \frac{a_{\mathrm{BH}}\, \mathrm{SB}_y}{b_{\mathrm{BH}} + \mathrm{SB}_y}\ \exp\!\left(\varepsilon_y - \tfrac{1}{2}\, b_y\, \sigma_r^2\right),$$

Module: `get_recruitment()`

where σ_r is the standard deviation of the deviations and b_y is a year-specific bias-adjustment scalar. The lognormal bias-correction term $\frac{1}{2} b_y \sigma_r^2$ is modulated by a bias-adjustment ramp that reduces the correction in years where recruitment is poorly informed by the data, following the approach used in SS3 (Methot & Wetzel, 2013). Recruitment deviations may be treated as fixed effects (penalized maximum likelihood) or as random effects.

2.2 Likelihood components

The total objective function is the sum of the data likelihoods, the recruitment-deviation penalty (or random-effect contribution), parameter priors, and the harvest-rate penalty. Each data likelihood is controlled by a switch in the data object, so that any combination of CPUE, length composition, and weight composition data may be included or excluded without altering the model code.

2.2.1 Abundance indices

Relative abundance is fitted with a log-normal likelihood. The CPUE component (`get_cpue_like()`) supports one or more standardized indices, each with its own catchability q, additional observation variance, a power parameter ω relating the index to biomass, and a cumulative effort-creep term κ. For an observation in index j, the predicted index is formed from the vulnerable abundance $\tilde{N}$ associated with that observation,

$$\log \hat{I} = \log q + \omega \, \log \tilde{N} + \log \kappa,$$

Module: `get_cpue_like()`

and is then mean-centred across the observations of each index before comparison with the data. Observation and process variability are combined in quadrature, $\sigma = \sqrt{\sigma_{\text{obs}}^2 + \tau^2}$, where σ_{obs} is the input standard error of the observation and τ is an estimable additional standard deviation, and the contribution to the negative log-likelihood is

$$-\log \mathcal{L} = -\sum_i \log \phi\Big(\log I_i \,\Big|\, \log \hat{I}_i, \, \sigma_i\Big),$$

with $\phi(\cdot)$ the normal density.

2.2.2 Composition data

Length and weight composition data are fitted with a choice of likelihoods selected by the corresponding switch: a multinomial, a Dirichlet, or a Dirichlet-multinomial likelihood. Predicted compositions are obtained by converting predicted catch-at-age $\hat{C}_{f,a}$ to a length basis through the PLA (and, for weight composition, an additional length-to-weight rebinning matrix $\mathbf{G}$):

$$\hat{p}_{f,\ell} \propto \sum_a P_{\ell a} \, \hat{C}_{f,a} \qquad \text{(length)}, \qquad \hat{p}_{f,k} \propto \sum_{\ell,a} G_{k\ell} \, P_{\ell a} \, \hat{C}_{f,a} \qquad \text{(weight)}.$$

Modules: `get_length_like()`, `get_weight_like()`

Predicted proportions are aggregated into the active bin range for the fishery, with the tails optionally accumulated into the minimum and maximum bins, then a small robustness constant is added and the vector is renormalized to sum to one. For observation i of fishery f with observed sample size n_i, the three likelihood options are

$$-\log \mathcal{L}_i = \begin{cases} -\log \text{Multinomial}(\mathbf{o}_i \mid \hat{\mathbf{p}}_i), & \text{multinomial}, \\ -\log \text{Dirichlet}(\mathbf{o}_i \mid \hat{\mathbf{p}}_i \, n_i \, e^{\tau_f}), & \text{Dirichlet}, \\ -\log \text{DirMult}(\mathbf{o}_i \mid n_i, \, \hat{\mathbf{p}}_i \, e^{\tau_f}), & \text{Dirichlet-multinomial}, \end{cases}$$

where $\mathbf{o}_i$ is the observed composition (counts for the multinomial and Dirichlet-multinomial, proportions for the Dirichlet) and τ_f is a fishery-specific log-scale variance-adjustment parameter that scales the effective sample size. The composition likelihood is evaluated as a batched matrix operation over the observations of each fishery rather than a per-observation loop, which reduces the size of the automatic-differentiation tape.

2.2.3 Priors and penalties

Prior information is incorporated through a flexible prior specification (`evaluate_priors()`) that applies parametric priors to named parameters, together with the recruitment-deviation contribution (`get_recruitment_prior()`), which by default is a normal density on the deviations with standard deviation σ_r. These contributions allow the model to be fitted either as a penalized maximum likelihood problem or as a Bayesian model (Section 3.0.2).

3 Features

3.0.1 Projections

`opal` supports catch-based forward projection of the population from an estimated set of parameters. Projections are conditioned on a specified future catch and propagate parameter uncertainty into the projected quantities. Two approaches to characterizing uncertainty in the projections are available: resampling from a multivariate normal approximation to the joint parameter distribution at the mode, and propagation of a posterior sample obtained by Markov chain Monte Carlo (MCMC; Section 3.0.2). Both approaches yield distributions of projected spawning biomass and depletion under the specified catch scenario.

3.0.2 Bayesian inference

Prior information can be incorporated either through a penalized maximum likelihood estimation (MLE) approach, in which priors enter the objective function as penalty terms and the model is optimized with `nlminb`, or through direct Bayesian inference. Bayesian estimation is performed with `SparseNUTS` (C. Monnahan, 2026), which implements a sparse-preconditioned variant of the No-U-Turn Sampler that exploits the sparse precision structure of TMB models to improve sampling efficiency for high-dimensional, correlated, and hierarchically structured posteriors (C. C. Monnahan et al., 2026). Because the `opal` objective function and its component modules are constructed in RTMB, the same model definition is used for both the MLE and the Bayesian workflows.

3.0.3 Model diagnostics

`opal` provides a set of built-in diagnostics intended to support model evaluation and validation. These include estimability checks and parameter-correlation summaries derived from the fitted model; simulation self-testing, in which data are simulated from the model and re-estimated to confirm that parameters can be recovered; posterior-predictive checking where the observations are compared to the simulated distributions of the predictions given the model and one-step-ahead (OSA) residuals, computed through RTMB's `oneStepPredict()`, which provide residuals that are independent and identically distributed under a correctly specified model and are therefore more readily interpretable than ordinary residuals for non-Gaussian observations (Kristensen et al., 2016). The simulation and posterior-predictive capabilities also support the projection and Bayesian workflows described above.

4 Case studies

Two case studies are distributed with the package as worked examples. The first applies `opal` to real data from an existing WCPFC tuna assessment; the second applies it to simulated data for which the underlying population dynamics are known. Detailed configuration, code, and outputs for each are provided in the package vignettes.

4.1 2020 WCPO bigeye tuna

The first case study uses the data from a fleets-as-areas model described in the appendix of the 2020 assessment of bigeye tuna (*Thunnus obesus*) in the western and central Pacific Ocean (Ducharme-Barth et al., 2020). The bundled data object contains the biological parameters, catch and standardized CPUE observations, length and weight composition data, configured at a quarterly time step across the fisheries defined in the fleets-as-areas assessment, with selectivity specified by fishery as either logistic or double-normal forms.

In its current configuration, this case study is implemented as an age-structured production model (ASPM): selectivity is held fixed at values converted from the original assessment, and the model is fitted to the standardized CPUE index only, with the size-composition likelihoods switched off. Model fit is evaluated using the OSA residuals of the CPUE index (Section 3.0.3). Fitting to the size-composition data within this configuration is an area of ongoing development (Section 5.1).

4.2 Hawai'i 'Opakapaka

The second case study uses simulated data for Hawai'i 'opakapaka (*Pristipomoides filamentosus*), a deepwater snapper, for which the true population dynamics are known. The reference model was constructed in SS3 (v3.30.19; Methot & Wetzel, 2013) as part of an associated simulation study (Oshima et al., 2026), which provides both an operating model that generates the data and an estimation model fitted to those data. The `opal` model covers the historical period 1949–2023 with three fleets — a commercial fishery, a non-commercial fishery, and a research-fishery survey — at an annual time step, and fits to two CPUE indices, generated with negligible observation error, together with length composition data from the commercial and survey fleets.

The case study is configured to estimate population scale, annual recruitment deviations, initial age structure, and selectivity parameters for the fleets that have length composition data, with the remaining life-history and fishery parameters fixed at the values used to generate the simulated data. Because the operating model is built in SS3, the case study also serves as a cross-platform comparison: it provides a basis for evaluating the correspondence between the `opal` and SS3 population-dynamics calculations under identical inputs, and for comparing the spawning-biomass trajectory and parameter estimates from `opal` against the operating-model truth and the SS3 estimation model.

Fitting the length composition likelihood and freeing selectivity for the commercial and survey fleets, the resulting estimated spawning-biomass trajectory is compared against the SS3 operating and estimation models in Figure 1. Model convergence is summarised by Table 1.

5 Discussion

`opal` provides an age-structured, length-based stock assessment framework implemented in RTMB, with a modular internal structure, support for multiple data types and likelihood forms, random-effects estimation, catch-based projections, Bayesian inference, and a set of built-in diagnostics. The two case studies demonstrate its application to both a real-data tuna assessment configured as an age-structured production model and a simulated single-species assessment that fits to index and length composition data and supports comparison against a known operating model.

5.1 Future development

`opal` is constructed as a set of standalone, individually documented module functions that are combined within the `opal_model()` objective function. Population-model components — growth, selectivity, recruitment, and the data likelihoods — are implemented as separate exported functions rather than embedded in a single routine, so that components can be modified, replaced, or extended independently, and so that individual modules can be tested in isolation. The collection of module functions is registered through a single accessor (`bet_globals()`) for use when the model is passed to the Bayesian sampler. This modular organization is intended to make the codebase tractable to maintain and to lower the barrier to community contribution.

Software development follows a defined contribution workflow. The package uses a two-branch structure: a `dev` branch for active development and a stable `main` branch for tested, release-ready code. New features are developed on branches forked from `dev` and are expected to include roxygen function documentation and minimal, self-contained `testthat` unit tests. Contributions are submitted as pull requests against `dev` , and are incorporated into `main` only after they pass the package checks — that is, after all tests pass and the package and its vignettes can be built.

Integrated testing operates at several levels. Unit tests verify the behaviour of individual modules; for example, the growth tests confirm that `get_growth()` returns the reference lengths at the reference ages and is monotonic, and that the weight, maturity, and PLA conversions behave as expected. Integration tests verify that the modules combine correctly within the full `opal_model()` objective function and that the objective and its gradient are finite when particular data and likelihood components are active. A numerical-regression baseline (`opal_baseline`), produced and checked through a dedicated vignette, stores a complete reference evaluation of the bigeye case-study model — the objective value, the gradient, the report output, and timing — so that a refactored build can be compared against the reference to detect unintended numerical changes and performance regressions. Continuous-integration checks (`R-CMD-check`) are run automatically on the repository.

Several extensions are planned. Priorities identified for ongoing work include refinement of the case studies; code optimization, particularly of the selectivity and size-composition likelihood calculations; and additional functionality such as biological and fishery reference points, alternative selectivity forms, and conditional age-at-length data. Within the WCPFC context, planned work also includes a side-by-side comparison of MULTIFAN-CL (Fournier et al., 1998), SS3 (Methot & Wetzel, 2013), and `opal` for a WCPO bigeye tuna fleets-as-areas model, an Indian Ocean yellowfin tuna fleets-as-areas case study based on simulated data, and potential application to the 2027 North Pacific blue shark stock assessment.

6 Acknowledgements

Development of opal is intended to be an open and collaborative effort, inclusive to the broader fisheries assessment community. To that end we are grateful to the many people that contributed to the project through informal discussions, and participation in periodic update meetings. While not an exhaustive list, this community includes: N. Davies (Te Takina Ltd), S. Hoyle (Hoyle Consulting), M. Maunder (Inter-American Tropical Tuna Commission), R. Bi (IATTC), F. Carvalho (PIFSC), G. Pilling (SPC), J. Hampton (SPC *emeritus*), and many others including participants in the 2025 CAPAM mini workshop on developing the next generation tuna stock assessment model as well as the 2026 SPC Pre-Assessment Workshop.

7 Declaration of generative AI use

Generative AI tools were used to assist with code development, refactoring, and documentation, and with drafting and editing text in this document. All analytical choices, model specifications, diagnostic interpretations, and conclusions are those of the authors, who reviewed and verified all AI-assisted output.

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9 Tables

Table 1: opal (with LF) convergence diagnostics: positive-definite Hessian, maximum absolute gradient (mgc), minimum Hessian eigenvalue, and parameter estimability.

Diagnostic	Value
Positive-definite Hessian	TRUE
Max |gradient| (mgc)	2.50e-03
Min Hessian eigenvalue	2.47e-01
Inestimable parameters	0 (all parameters estimable)

10 Figures

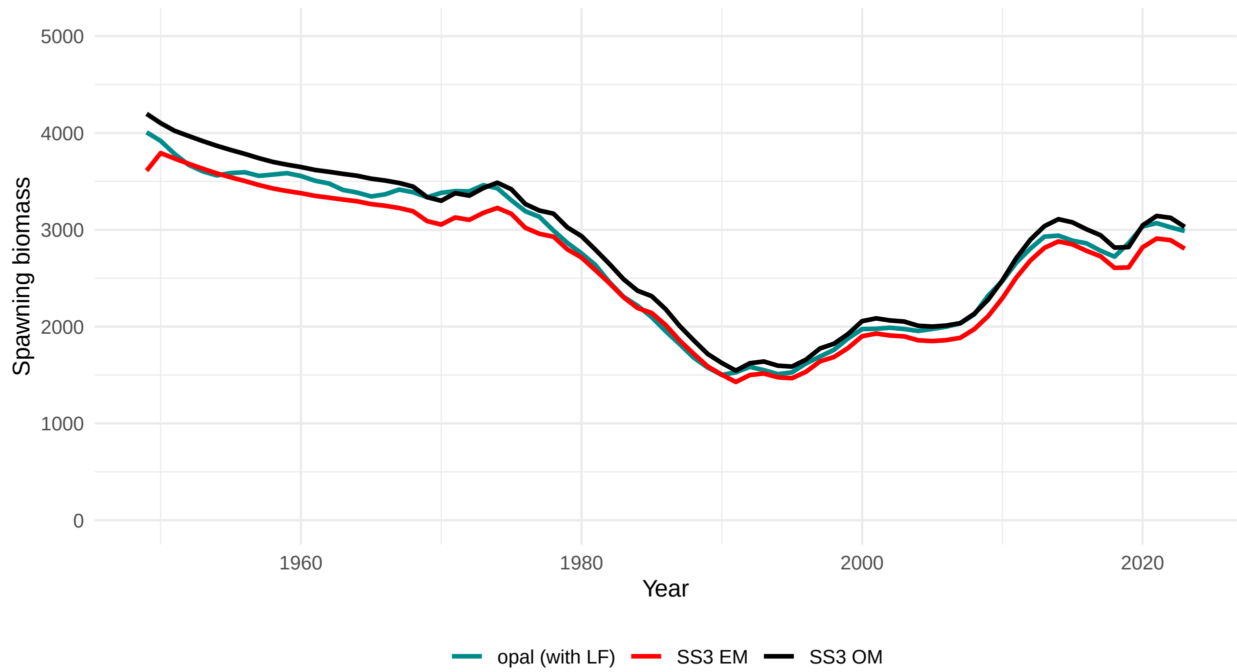

Figure 1: Estimated spawning-biomass trajectory for the 'opakapaka opal (with LF) model compared with the SS3 operating model (OM) and estimation model (EM), 1949–2023.